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CS4420

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Assignment 5

The program, assignment5.cc is compiled by running “g++ assignment5.cc”. To use the program, use “./a.out inputfilename [FCFS|SJF|RR] [timequantum]”. FCFS for First come first serve, SJF for shortest job first, and RR for round robin scheduling algorithms respectively. Timequantum is only used with the round-robin algorithm, and trying to run the round robin algorithm without a positive integer time quantum (or no time quantum) will print an error message and exit the program. Likewise, not specifying a scheduling algorithm or an input file name (or an invalid input file name) will also print an error message and end the program. Input.1, input.3, and input.4 are all valid input files for the program, while input.2 is an invalid input file used to check that the program will stop executing upon reading a file with an invalid number of processes specified in the file. Input.1 uses the input data of the sample input from the assignment.

Example compilation and program running with input.1 with each scheduling algorithm:

(normal test case 1)

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ ls
a.out  assignment5.cc  input.1  input.2
dinoguts@Lclasswork25:~/cs4420/assignment5$ g++ assignment5.cc
dinoguts@Lclasswork25:~/cs4420/assignment5$ ./a.out input.1 FCFS
```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	12	12	0
1	2	12	16	4	10
2	3	16	17	1	13
3	4	17	19	2	13

Average waiting Time: 9

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ █
```

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ ./a.out input.1 SJF
```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	12	12	0
2	3	12	13	1	9
3	4	13	15	2	9
1	2	15	19	4	13

Average Waiting Time: 7.75

```
dinoguts@Lclasswork25:~/cs4420/assignment5$
```

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ ./a.out input.1 RR 5
```

PID	Start Time	End Time	Running Time
0	0	5	5
1	5	9	4
2	9	10	1
3	10	12	2
0	12	17	5
0	17	19	2

PID	Arrival Time	Running Time	End Time	Waiting Time
0	0	12	19	7
1	2	4	9	3
2	3	1	10	6
3	4	2	12	6

Average Waiting Time: 5.5

```
dinoguts@Lclasswork25:~/cs4420/assignment5$
```

Input.3 data:

```
3
0 1 5
1 2 3
2 10 4
```

Each algorithm running with input.3 data:

(normal test case 2)

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ ./a.out input.3 FCFS
```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
Idle	-	0	1	1	
0	1	1	6	5	0
1	2	6	9	3	4
Idle	-	9	10	1	
2	10	10	14	4	0

Average waiting Time: 1.33333

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ ./a.out input.3 SJF
```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
Idle	-	0	1	1	
0	1	1	6	5	0
1	2	6	9	3	4
Idle	-	9	10	1	
2	10	10	14	4	0

Average Waiting Time: 1.33333

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ ./a.out input.3 RR 5
```

PID	Start Time	End Time	Running Time
Idle	0	1	1
0	1	6	5
1	6	9	3
Idle	9	10	1
2	10	14	4

PID	Arrival Time	Running Time	End Time	Waiting Time
0	1	5	6	0
1	2	3	9	4
2	10	4	14	0

Average Waiting Time: 1.33333

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ █
```

Input.4 data

```
2
0 0 4
1 2 3
```

Each algorithm running with input.4 data:

(normal test case 3)

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ ./a.out input.4 FCFS
```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	4	4	0
1	2	4	7	3	2

Average waiting Time: 1

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ ./a.out input.4 SJF
```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	4	4	0
1	2	4	7	3	2

Average Waiting Time: 1

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ ./a.out input.4 RR 2
```

PID	Start Time	End Time	Running Time
0	0	2	2
1	2	4	2
0	4	6	2
1	6	7	1

PID	Arrival Time	Running Time	End Time	Waiting Time
0	0	4	6	2
1	2	3	7	2

Average Waiting Time: 2

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ █
```

Error handling test 1, invalid time quantum:

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ ./a.out input.1 RR -4
```

Invalid time quantum

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ █
```

Error handling test 2, missing time quantum:

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ ./a.out input.1 RR
```

Error. Time quantum not found

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ █
```

Error handling test 3, invalid number of arguments:

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ ./a.out input.1  
Invalid number of arguments  
dinoguts@Lclasswork25:~/cs4420/assignment5$ █
```

Error handling test 4, invalid file name:

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ ./a.out input.1 SJF  
Could not open file  
dinoguts@Lclasswork25:~/cs4420/assignment5$ █
```

Error handling test 5, invalid number of processes specified in file:

```
dinoguts@Lclasswork25:~/cs4420/assignment5$ ./a.out input.2 SJF  
Invalid number of processes  
dinoguts@Lclasswork25:~/cs4420/assignment5$ █
```